Human–Machine Cocreation: The Effects of ChatGPT on Students' Learning Performance, AI Awareness, Critical Thinking, and Cognitive Load in a STEM Course Toward Entrepreneurship

Yu Ji, Zehui Zhan, Tingting Li, Xuanxuan Zou, and Siyuan Lyu

***Abstract*—The advent of generative artificial intelligence (GAI), exemplified by ChatGPT, has introduced both new opportunities and challenges in science, technology, engineering, and mathematics (STEM) and entrepreneurship education. This exploratory quasi-experimental study examined the effects of ChatGPT-assisted collaborative learning (CCL) on students' learning performance, artificial intelligence (AI) awareness, critical thinking, and cognitive load. A total of 36 sophomore undergraduates participated in an eight-week instructional experiment, dedicating 3 h per week to applying STEM and entrepreneurship knowledge in the creation of cultural products. The experimental group ($N = 21$) participated in CCL, while the control group ($N = 15$) engaged in non-ChatGPT-assisted collaborative learning (NCCL). The results indicated that the CCL group outperformed the NCCL group in terms of learning performance, AI awareness, and cognitive load, while the NCCL group excelled in critical thinking. The findings confirm that ChatGPT offers significant potential and advantages in addressing complex problems within group collaboration and stimulating group creativity, providing new insights into fostering students' entrepreneurial spirit and skills. However, overreliance on and misuse of ChatGPT may hinder student learning outcomes. Future research should focus on the cocreative problem-solving mechanisms between humans and machines in entrepreneurial education, particularly the interplay of knowledge, thinking, emotions, and actions in collaborative processes involving GAI.**

***Index Terms*—Artificial intelligence (AI) awareness, ChatGPT, cognitive load, critical thinking, learning performance, STEM and entrepreneurship education.**

I. Introduction

A NEW wave of technological and industrial transformation is emerging, attracting increasing attention to science, technology, engineering, and mathematics (STEM) and entrepreneurship education [1]. STEM education and entrepreneurship education have a deep and complementary relationship. The integration of these two fields not only enhances the practical value of STEM education but also adds essential technical depth and innovative potential to entrepreneurship education [2]. In essence, entrepreneurship education centers on turning creative ideas into business opportunities, whereas STEM education, by fostering applied knowledge, helps students understand how to convert scientific and technological innovations into products and services that market demands. Thus, STEM education equips students with an entrepreneurial mindset, enabling them to identify opportunities, assess risks, generate solutions, and create products for their future entrepreneurial ventures. For example, identifying and solving complex problems requires students to apply and integrate knowledge across different domains, recognize market needs through unique perspectives, address business challenges, propose innovative solutions, and generate product value [3]. Studies have demonstrated that STEM education can foster students' entrepreneurial behavior, intentions, and spirit, cultivating the skills and adaptability necessary in a rapidly changing global market [4], [5].

One of the key trends in the development of STEM education is the integration of intelligent technologies. The adoption of artificial intelligence (AI) technologies or products can enhance students' entrepreneurial skills and thinking [6]. Natural language processing (NLP) models like ChatGPT, which use reinforcement learning from human feedback, can perform complex tasks with enhanced reliability and creativity [7]. By emulating human learning and thinking, it helps students deepen their understanding of complex STEM concepts [8], offers real-time entrepreneurial guidance and market analysis [9], and accelerates the divergence and convergence of individual and group thinking [10]. However, the application of technology is a double-edged sword: excessive reliance on AI may lead to superficial cognitive thinking engagement, diminished teacher–student interaction, emotional disengagement, and cognitive dissonance, all of which hinder learners' active construction of meaning [11]. Therefore, both teachers and students must possess the requisite AI literacy and ethical standards to address potential risks in the human–machine cocreation process and develop appropriate strategies to integrate ChatGPT into course learning.

Received 5 February 2024; revised 27 May 2024, 3 December 2024, 21 February 2025, and 26 February 2025; accepted 21 March 2025. Date of publication 26 March 2025; date of current version 24 April 2025. This work was supported by the 2024 Guangdong Province Undergraduate Universities Teaching Quality and Teaching Reform Project: Multi-Agent Empowered "Cross-Innovation Integration" Talent Cultivation Model and Practical Research. *(Corresponding author: Zehui Zhan.)*

The authors are with the School of Information Technology in Education, South China Normal University, Guangzhou 510631, China (e-mail: 1538117109@qq.com; zhanzehui@m.scnu.edu.cn; lttsicnu@163.com; zxx200010@163.com; 2021020889@m.scnu.edu.cn).

Digital Object Identifier 10.1109/TLT.2025.3554584

Collaboration is recognized as a critical means to harness team creativity in STEM courses [12]. Students collaboratively create value for business products through market analysis, problem exploration, solution design, prototype iteration, and value promotion. Throughout the entire process, ChatGPT takes on the role of a "quasi-member" in collaborative learning, working alongside team members to solve complex problems. However, the introduction of ChatGPT may alter how students process information and solve problems, thereby impacting their learning performance [13], AI awareness [14], critical thinking, and cognitive load [11]. Despite the extensive empirical research on ChatGPT in core subjects such as mathematics and English, there has been comparatively little focus on its role in STEM courses aimed at fostering entrepreneurship [9]. ChatGPT's influence as a learning partner or expert in learning STEM and entrepreneurship concepts, as well as in the creation of cultural products, remains unclear. This study will focus on assessing the specific roles and effectiveness of ChatGPT in cultural product creation through collaborative group work. Furthermore, contemporary human–machine collaborative learning mainly focuses on one-on-one interactions between humans and machines [11], while scenarios involving multiple users and machines remain relatively underexplored. This study will focus on multiuser multimachine environments, leveraging human–machine collaboration to address problems and create cultural products, with the aim of investigating the pathways and practical impacts of such collaborative efforts.

II. Literature Review

In the AI era, intelligent technologies have become ubiquitously integrated in people's lives and learning. The advancement of AI has redefined human–machine collaboration, with the application of ChatGPT in STEM courses toward entrepreneurship significantly influencing students' collaboration, thinking, and cognition. Therefore, this study will conduct a literature review focusing on the aforementioned two aspects.

A. Human–Machine Collaborative Learning

The incorporation of ChatGPT shifts "human–human" interaction to a combined "human–machine–human" collaborative approach, characterized by shared cognition, perception, thinking, and decision making between humans and machines [15].

Human–machine collaborative learning originates from synergetic theory, which suggests that every component in a system adapts and evolves through ongoing interactions, thereby influencing the state and behavior of other system elements [16]. Intelligent machines can take on multiple roles in group interactions, providing diverse perspectives and information resources, thereby facilitating meaning negotiation and the co-construction of knowledge. Furthermore, distributed cognition theory underpins the theoretical framework of human–machine collaborative learning. This theory posits that cognition extends beyond the individual mind, being distributed across an entire ecosystem comprising people, culture, technology, and artifacts [17]. Serving as an extended cognitive tool, intelligent machines can provide various cognitive resources in group collaborations, alleviating cognitive load during information processing and enabling students to focus more on creative tasks. As AI technology advances, representative teaching robots [18] and intelligent agents [19] have increasingly evolved into quasi-subjects. They actively engage in communication and interaction with learners within learning communities, facilitating information exchange, stimulating ideas, and providing timely evaluation [20]. Within the collaborative group, students and intelligent machines cocreate a network for meaning construction, fostering the integration of human and machine intelligence. The artifacts collaboratively produced by humans and machines serve as crucial elements of cognition, embodying the externalization of learners' knowledge and thought processes [21]. In essence, these technologies are designed to provide humanoid learning companions that facilitate natural interactions and collaboration between learners and these robots or agents, ultimately enhancing learning effectiveness and quality.

In practical settings, traditional collaborative learning often faces challenges, such as inadequate guidance, limited group interaction, a narrow approach to solving complex problems, and difficulties in integrating knowledge across different disciplines [22], [23]. Synergetic theory and distributed cognition theory provide robust frameworks for addressing these challenges. Research has showed that chatbots and other intelligent support systems can significantly improve students' learning experiences [24]. By mimicking the role of human instructors, these systems deliver real-time tailored feedback and guidance, enabling students to engage more deeply with learning material. For example, chatbots can facilitate students' exploration of complex concepts through iterative dialogues, fostering deeper insights and understanding through continuous questioning and negotiation [25]. NLP tools are employed to assess students' writing, offering feedback on grammar, spelling, and various writing skills [26]. In collaborative projects, ChatGPT can collect and structure information according to students' requirements, facilitating brainstorming and idea generation, which in turn boosts the efficiency and creativity of problem solving [27]. In addition, intelligent tutoring agents deliver personalized feedback and diagnostics on specific content, playing a vital role in enhancing critical tasks like business plan development or product presentation preparation [28]. Consequently, AI systems with human-like characteristics act as "quasi-partners" in collaborative learning and problem-solving scenarios, assisting team members in crafting artificial artifacts.

However, in human–machine collaboration, relationships dominated by human control tend to be more reliable [29]. When students struggle to maintain clear and independent awareness in using technology, they may easily succumb to "technology obsession" and "technology worship" [30], shifting from active learners to passive executors of machine directives, which can severely hinder their independent thinking abilities. Furthermore, the inherent "bias," "discrimination," and "preset" characteristics of technology can lead to ethical dilemmas, including false information identification, digital privacy breaches, and compromised academic integrity [31]. Thus, in practical teaching, it is essential to help students understand the functions, features, advantages, and limitations of generative artificial intelligence (GAI) tools. Instruction should focus on cognitively challenging tasks, encouraging students to leverage machine intelligence to create meaningful cultural artifacts that respond societal needs.

B. Application of ChatGPT in STEM Courses Toward Entrepreneurship

ChatGPT, a sophisticated AI content generation tool, leverages NLP and advanced machine learning algorithms, incorporating 175 billion parameters [32]. By mimicking human emotions and tones, it engages in user interactions in a manner akin to human reasoning, offering contextually relevant responses based on user queries. ChatGPT serves as a crucial intermediary for information exchange between learners and their learning environment. It facilitates multiperspective and comprehensive analysis of the real-world context, which often contains contradictory information, thereby enabling students to solve complex problems more efficiently [33]. Specifically, ChatGPT can enhance students' learning by offering tailored support such as demand-based customization, real-time interaction, innovative feedback, creative inspiration, information synthesis, and personalized guidance [34].

Integrating ChatGPT into STEM courses toward entrepreneurship represents an innovative initiative aimed at fostering creative ideas, and the rapid prototyping of business models [35]. This integration not only enhances students' learning performance but also influences their AI awareness, critical thinking abilities, and cognitive load. For instance, by simulating realistic business communication scenarios, ChatGPT assists in analyzing diverse information sources, trends, and markets shifts, effectively identifying potential business opportunities or favorable conditions for cultural and creative products [9], [36]. Students can engage in drafting and revising business plans through iterative interactions with ChatGPT, enhancing the value, functionality, and originality of artificial artifacts. In addition, incorporating ChatGPT is pivotal for fostering students' AI awareness. Through their interactions with ChatGPT, students not only learn to leverage AI in solving practical problems, but also develop a nuanced understanding of AI's mechanisms and its potential applications in business [37]. Individuals with heightened AI awareness are more inclined to utilized AI tools for acquiring knowledge and solving problems, believing that intelligent machines enhance personal productivity and learning performance [38].

In entrepreneurship education, critical thinking serves as a cornerstone of students' success. ChatGPT facilitates the development of this skill by offering diverse solutions and perspectives, compelling students to critically assess the feasibility and effectiveness of AI-generated proposals [35]. During human–machine interactions, learners must actively evaluate the accuracy and relevance of information to ensure that it aligns with potential solutions [39]. Nevertheless, excessive reliance on or improper use of ChatGPT may impede the cultivation of students' critical thinking skills [40]. Furthermore, leveraging its extensive training on diverse datasets to understand and interpret human knowledge, ChatGPT can reorganize knowledge content by synthesizing complex information. It provides expert-like explanations of STEM or entrepreneurship concepts, thereby deepening learners' comprehension of these subjects [13]. Studies indicate that ChatGPT significantly alleviates the cognitive associated with acquiring and understanding knowledge [41]. Similarly, research demonstrates that ChatGPT reduces the cognitive load of problem exploration by offering learners problem clues and constructing learning scaffolds [42].

In a technology-driven learning environment, GAI plays a pivotal role in supporting students' acquisition of entrepreneurial knowledge and participation in STEM innovation practices. Students must address the profound influence of GAI tools on business creation, operation, product design, and development. Integrating these emerging technologies into educational settings is essential to fostering creativity and unlocking students' imaginative potential.

C. Research Hypotheses

The reviewed literature suggests that human–machine collaborative learning is poised to be a transformative approach in future STEM classrooms. GAI tools, such as ChatGPT, exhibit advanced "capabilities" and "intelligence" that support students in entrepreneurial training, fundamentally altering the traditional dynamics between humans and machines. While synergetic theory and distributed cognition theory provide clear advantages for students in STEM courses, research remains limited on its sustained impacts, especially concerning the influence of GAI tools on learners.

Furthermore, prior research has predominantly conceptualized ChatGPT as an AI chat tool designed for individual one-on-one interactions with learners [11]. Its application programming interface (API) currently lacks the functionality to integrate into social groups, preventing multiturn dialogues involving multiple users. To address this underexplored yet compelling dimension, this study investigates the efficacy of ChatGPT in STEM entrepreneurship courses, focusing on key aspects such as learning performance, AI awareness, critical thinking, and cognitive load. This study proposes the following four research hypotheses.

1) Integrating ChatGPT into STEM courses toward entrepreneurship can enhance students' learning performance.
2) Integrating ChatGPT into STEM courses toward entrepreneurship can foster students' AI awareness.
3) Integrating ChatGPT into STEM courses toward entrepreneurship can support the development of students' critical thinking.
4) Integrating ChatGPT into STEM courses toward entrepreneurship can alleviate students' cognitive load.

III. Materials and Methods

A. Participants

This study was carried out within a STEM course toward entrepreneurship at a public normal university in southeast China. The course spanned 16 weeks, with 3 h of instruction per week. The participants comprised 36 students (mean age: 19.97 years) and a teaching team consisting of one professor and three teaching assistants). Before the course began, we found that most students had prior experience with AI tools for learning, including speech recognition, image recognition, chatbots, and similar application. Among these students, 80.55% reported using AI tools between zero and two times per week. A total of 94.44% of students reported that they have never participated in competitions or knowledge-based activities related to programming, robotics, or AI. Only 14.30% of students reported that they have used GAI. This indicates that while most students have experience using AI tools, they lack basic knowledge and skills related to GAI.

During the first week of the course, the instructor informed students that after eight weeks of theoretical learning, they would be required to create a business product and promote its value during the last eight weeks. In response, students expressed a preference for engaging in group collaboration. Subsequently, following an icebreaker activity, the 36 students formed groups and elected leaders based on their abilities and strengths. Prior to the experimental phase (eighth week), a teaching assistant informed students that certain groups would utilize ChatGPT for cultural product creation from weeks 9–16 and collected group preferences. To mitigate potential selection bias, group leaders randomly determined through a lottery whether their group would use ChatGPT. After the drawing, one group expressed a desire to switch roles and not use ChatGPT, while another group, comprising three members with prior experience of using chatbots, expressed a stronger preference for incorporating ChatGPT. Following a negotiation, the two groups agreed to switch roles. Ultimately, 21 students (2 males and 19 females) were assigned to the ChatGPT-assisted collaborative learning (CCL) group. These students were divided into four subgroups: one consisting of six members and three with five members each. The remaining 15 students (9 males and 6 females) were assigned to non-ChatGPT-assisted collaborative learning (NCCL) group, which was divided into three groups of five students each. Participants attended this STEM course toward entrepreneurship weekly, but they had the flexibility to schedule lab sessions to complete their cultural products. In addition, baseline assessments of AI awareness, critical thinking, and cognitive load indicated comparable levels between the two groups.

B. Procedure

In the eighth week, the teaching team integrated the ChatGPT-3.5 API into the CCL group's WeChat group. Within this setup, WiseGPT facilitated continuous discussions on a single topic or question among multiple group members by comprehending the contextual content and responding to inquiries interactively. GPT provided individualized one-on-one services to students. Students could mention (@) GPT to request exclusive Q&A sessions if their specific questions or needs were not addressed, as shown in Fig. 1. For instance, Student 1 posed the question to WiseGPT, "I need to make a wooden dragon boat model that can operate on water and carry a certain volume of modules. What aspects should I consider in the design of this model?" Building on this, Student 2 added an identity and tool supplement, asking, "As a model designer, please list the precautions for making design drawings in Laser maker."

Fig. 1. Human–machine collaborative dialogue for problem solving.

Second, teaching assistant A dedicated approximately 90 min to introducing the CCL group students to the functionalities of ChatGPT. This session included instructions on prompt commands, text composition, data analysis, and intelligent search, followed by 20 min for students to engage in hands-on interaction with ChatGPT. ChatGPT actively participated in the group collaborative learning process by performing various roles: answering questions (e.g., providing insights on cultural, programming, and entrepreneurial knowledge), clarifying perspectives (resolving debates within the group), organizing information (e.g., synthesizing literature and customer interview date), processing data (e.g., market analysis and enterprise operations data), providing feedback (e.g., evaluations of business plans and cultural products), and diagnosing solutions (e.g., offering suggestions and recommendations for problem-solving strategies). For example, students in the CCL group utilized ChatGPT for tasks, such as market analysis, identifying the entrepreneurial value and potential of cultural products, refining problem-solving strategies, and reducing time costs. To maintain experimental integrity, this course content was intentionally withheld from the NCCL group. Meanwhile, teaching assistant B presented the NCCL group with examples of business products created by students in previous cohorts and invited accomplished senior students to share experiences, including strategies for identifying business opportunities and creating produce business products. During the formal experiment, the NCCL group adhered to the traditional group collaboration method, involving task division. They accessed entrepreneurial information and solved complex problems through platforms like Baidu,[1] Zhihu,[2] and Bilibili.[3]

The primary objective of offering this STEM course toward entrepreneurship is to equip students with fundamental entrepreneurial methods and foster their ability to apply interdisciplinary thinking to address entrepreneurial challenges. In addition, the course encourages students to engage in hands-on production of cultural products and their value promotion. Given that students had no prior experience with STEM courses toward

[1] [Online]. Available: https://www.baidu.com/
[2] [Online]. Available: https://www.zhihu.com/
[3] [Online]. Available: https://www.bilibili.com/

TABLE I
COURSE PLAN

Stage	Week	Contents
Theoretical knowledge and tools learning	First week	**Entrepreneurial Spirit and Opportunity Recognition:** Exploration of entrepreneurship and entrepreneurial spirit; STEM disciplines as drivers of innovation and opportunity discovery; methodologies for assessing and evaluating opportunities; pretest questionnaire implementation.
	Second week	**From Ideas to Prototypes:** Application of creative thinking tools; core principles of the Minimum Viable Product (MVP); feedback integration for iterative development.
	Third week	**Business Models and Financial Planning:** Utilizing the Business Model Canvas to develop business plans; understanding fundamental financial concepts.
	Fourth week	**Marketing Strategies and Brand Building:** Foundational marketing theories and strategies; leveraging digital marketing and social media; creating and sustaining a compelling brand image.
	Fifth week	**Investor Relations and Presentation Skills:** Crafting persuasive investor pitches; identifying financing channels and strategies; defining presentation and negotiation skills.
	Sixth week-to Seventh week	**Learning STEM Tools:** Learning and practicing mind+ software, open-source hardware, laser cutting, and 3D printing.
	Eighth week	**Integrating API:** The ChatGPT-3.5 API was integrated into the CCL Group's WeChat group, and its functionalities and usage were systematically explored.
STEM entrepreneurship education project practice	Nineth week	**Task 1:** Market Analysis and Problem Exploration: Conduct a comprehensive analysis of the cultural product market; identify potential product concepts; pinpoint challenges and obstacles in cultural product development.
	Tenth week	**Task 2:** Solution Design: Identify and segment target customer groups; develop tailored business plans for the cultural products of each group.
	11th week to 14th week	**Task 3:** Prototype Iteration: Employ diverse software and hardware tools to create cultural product prototypes; conduct multidimensional testing, assess user needs; refine products for continuous enhancement.
	15th week to 16th week	**Task 4:** Value Promotion: Conduct business plans roadshow simulations; compose reflective essays on learning experiences; finish post test questionnaires.

entrepreneurship, the curriculum was structured as follows (see Table I). During weeks 1–8, students acquired foundational knowledge in entrepreneurship, as well as software and hardware applications in STEM. In weeks 9–16, students participated in STEM-based entrepreneurial activities, progressing through stages such as market analysis, problem exploration, solution design, prototype iteration, and value promotion.

Students completed a 15-min questionnaire pretest on the first day and a posttest on the last day of course. At the conclusion of each weekly session, students submitted group-based learning reflections, highlighting challenges encountered and reporting task progress. At the end of the course, each student submitted an individual learning reflection report. These reflections informed ongoing course optimization and allowed for more targeted learning support. Both groups engaged with identical learning content and were tasked with creating cultural products. During the value promotion stage, a panel comprising four master's students in educational technology, three doctoral students, and one professor, all with similar evaluation experience, conducted sympathetic evaluations of the cultural products.

C. Materials

The course materials included the following.

1) *Laser maker software and its accompanying laser cutter:* Students began by designing their cultural products using Laser maker software. Then, they used the laser cutter to cut wood or acrylic panels, enabling the physical assembly of their cultural products.
2) *Mind+ software, combination with mainstream control boards and diverse open-source hardware components:* The functionalities of the cultural products were achieved using open-source hardware and programming. For example, students enabled a dragon boat to move forward and developed a feedback system for answering questions.
3) *Task sheets:* The CCL group's task sheets included details such as the time, issues the group aimed to address, ChatGPT's responses to those issues, the intent and method of inquiry, and the outcomes (resolved or unresolved). In contrast, the NCCL group's task sheets included details, such as the time, issues the group aimed to resolve, key topics and content of group discussions, task distribution, and outcomes (resolved or unresolved). These sheets documented the group's approach to addressing challenges during the creation of cultural products, both with and without the assistance of ChatGPT.

D. Measure Instrument

To assess the effectiveness of ChatGPT in supporting students' creation of cultural products, this study utilizes measurement tools to evaluate its impact on four dependent variables: learning performance, AI awareness, critical thinking, and cognitive load.

1) *Learning performance:* Since creativity is a critical skill for entrepreneurs in developing innovative products and services [43], this STEM course toward entrepreneurship emphasizes product creation, encouraging students to identify business opportunities through market analysis and produce innovative and valuable cultural products aligned with user needs. To evaluate the creativity of the products, we employed widely accepted assessment techniques to score each group's product based on novelty, fluency, and originality [44], [45]. Novelty evaluates whether the presented ideas are distinct or innovative, incorporating uniqueness and fresh perspectives. Fluency examines the quantity, coherence, and logical consistency of ideas. Originality pertains to exploring materials to generate novel and broadly accepted outputs.
2) *AI awareness:* To evaluate the changes in students' AI awareness before and after the course, we adopted two dimensions from [25], which demonstrated acceptable reliability in comparable age groups. If students recognize the benefits of ChatGPT in product creation, they are likely to increase both the frequency and efficiency of its use in future learning, thereby enhancing the quality of their

problem solving. AI awareness is categorized into two dimensions: perceived comfort ($\alpha = 0.89$) and attitudes toward learning with ChatGPT ($\alpha = 0.87$). Each dimension was assessed using four items on a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

3) *Critical thinking:* Students frequently face ill-structured complex problems alongside diverse and intricate information in this course. Students must identify and process relevant information during collaboration, evaluate its reliability and source, critically question perspectives—whether their own or those generated by AI—and reason from multiple viewpoints to uncover new potential solutions. Consequently, this study utilized a questionnaire from [46] to evaluate students' critical thinking skills across five dimensions: truth seeking (three items, $\alpha = 0.830$), analyticity (three items, $\alpha = 0.632$), systematicity (eight items, $\alpha = 0.836$), inquisitiveness (seven items, $\alpha = 0.826$), and open-mindedness (seven items, $\alpha = 0.887$). Each dimension was assessed with a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

4) *Cognitive load:* According to the framework outlined in [47], cognitive load can be evaluated through the measurement of mental load, mental effort, and performance. This rating scale method relies on the fundamental assumption that students are capable of reflecting on their cognitive processes and reporting the mental effort exerted. Although single-item self-assessment scales might appear unreliable, research demonstrates that individuals possess sufficient capability to assign numerical values reflecting their perceived mental burden [48]. Therefore, students were instructed to assess the mental effort they invested both before and after the STEM entrepreneurial practice activities [49], using a nine-point scale ranging from 1 (very simple, requiring minimal cognitive effort) to 9 (very difficult, requiring maximum cognitive effort). For instance, students were prompted with the question, "How would you rate the mental effort you invested in STEM entrepreneurial practice activities?"

IV. Results

The data were analyzed using SPSS 25.0. In the first step, we examined the normality of the dependent variables, namely learning performance, AI awareness, critical thinking, and cognitive load. The results indicated that the pretest and posttest data for both groups of students followed a normal distribution in learning performance, AI awareness, and critical thinking dimensions ($p > 0.05$, according to the Shapiro–Wilk test), whereas the cognitive load dimension did not follow a normal distribution ($p < 0.05$).

In the second step, we performed statistical analyses on the pretest and posttest data for the learning performance, AI awareness, and critical thinking dimensions using paired samples t-test. In addition, we compared the posttest data between the CCL group and the NCCL group using independent samples t-test to evaluate the impact of ChatGPT on students during the creation of cultural products. For the cognitive load dimension, we analyzed the pretest and posttest data using the paired Wilcoxon test. The Mann–Whitney U test was employed as the nonparametric test to assess the statistical differences between the CCL group and the NCCL group.

In addition, we conducted a systematic review of the course's reflective reports and chat records, performing a detailed analysis. The findings from the qualitative analysis provide deeper insights and enhance the understanding of the results across the four variables.

TABLE II
COMPARISON OF LEARNING PERFORMANCE SCORE BETWEEN CCL AND NCCL GROUPS

Variable	Group	Sample (N)	Mean	SD	t	p
Learning performance	CCL	21	85.00	4.243	5.610	0.002
	NCCL	15	68.67	3.055		
Novelty	CCL	21	27.00	2.160	5.477	0.003
	NCCL	15	20.00	.000		
Fluency	CCL	21	35.00	1.826	5.861	0.002
	NCCL	15	27.33	1.528		
Originality	CCL	21	23.00	0.816	1.890	0.117
	NCCL	15	21.33	1.528		

A. Learning Performance

The learning performance results, presented in Table II, highlight notable differences between the CCL group and the NCCL group. The CCL group ($M = 85.00$, SD $= 4.243$) demonstrated a significantly higher overall mean score for learning performance compared to the NCCL group ($M = 68.67$, SD $= 3.055$), $t = 5.610$, $p = 0.002$. Regarding the subdimensions of novelty, the CCL group's mean score ($M = 27.00$, SD $= 2.160$) surpassed that of the NCCL group ($M = 20.00$, SD $= 0.000$), $t = 5.477$, $p = 0.003$, indicating a statistically significant difference. Students in the CCL group noted in their reflective reports that "ChatGPT can rapidly generate diverse ideas and inspiration for cultural products creation." Similarly, in the fluency subdimension, the CCL group achieved a significantly higher mean score ($M = 35.00$, SD $= 1.826$) compared to the NCCL group ($M = 27.33$, SD $= 1.528$), $t = 5.861$, $p = 0.002$. Chat records from the CCL group revealed discussions on topics such as "How to preserve and innovate the culture of the Dragon Boat Festival," "The commercial prospects and market value of dragon boat cultural and creative products," and "Approaches to designing and creating a dragon boat." Students in the CCL group participated in iterative discussions on these topics, fostering mutual inspiration and enhancing the continuity of their ideas. Regarding originality, the CCL group achieved a higher mean score ($M = 23.00$, SD $= 0.816$) compared to the NCCL group ($M = 21.33$, SD $= 1.528$), $t = 1.890$, $p = 0.117$, but this difference was not statistically significant. Interactions with the CCL group indicated that "the most queries focused on understanding concepts and implementation, primarily addressing improvements to existing products,

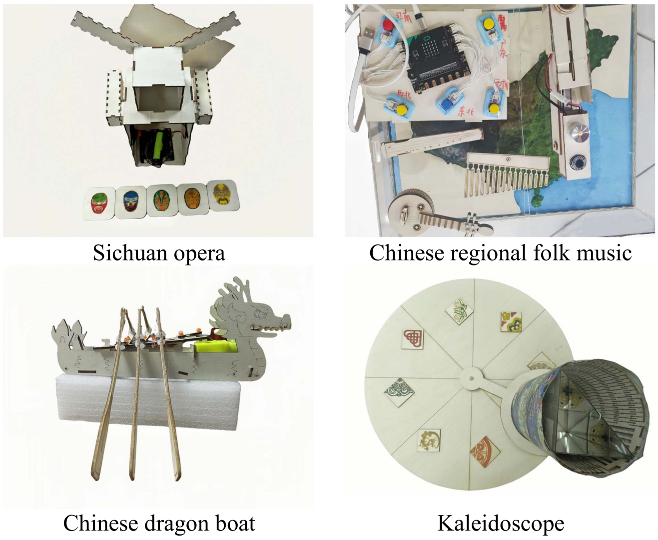

Fig. 2. Examples of cultural products created by students.

thereby constraining originality." Fig. 2 displays selected images of the cultural products created by students.

B. AI Awareness

The paired-samples t-test showed that the CCL group exhibited significant differences between pretest and posttest scores in AI awareness (t = –4.316, p = 0.000), as well as in its subdimensions of perceived comfort (t = –11.69, p = 0.000) and attitude (t = –6.388, p = 0.000), reflecting a positive trend in AI awareness development. In contrast, the NCCL group showed significant differences in AI awareness (t = –4.141, p = 0.001) and its subdimension perceived comfort (t = –2.955, p = 0.010), but no significant difference in attitude (t = –1.388, p = 0.187).

In addition, the posttest overall AI awareness score for the CCL group (M = 34.33, SD = 2.652) was significantly higher than that of the NCCL group (M = 32.47, SD = 1.685), t = 7.491, p = 0.000, indicating a statistically significant difference. Within the subdimensions, the perceived comfort mean score in the CCL group (M = 18.00, SD = 1.265) was significantly higher than that in the NCCL group (M = 15.47, SD = 0.915), t = 6.607, p = 0.000. Students noted that "the immediacy and generative capabilities of ChatGPT enable instant access to desired knowledge, providing a highly convenient and comfortable experience." Furthermore, the mean score for attitude in the CCL group (M = 18.57, SD = 1.660) was significantly higher than that in the NCCL group (M = 16.47, SD = 0.915), t = 4.865, p = 0.000 (see Table III). One student acknowledged ChatGPT's role in their work, stating, "ChatGPT suggested automating Sichuan Opera face-changing cultural and creative products through programming, which inspired our subsequent design and production efforts."

C. Critical Thinking

As shown in Table IV, the CCL group exhibited significant differences in overall critical thinking scores and its subdimensions ($p < 0.05$), with the exception of the subdimension systematicity

TABLE III
COMPARISON OF AI AWARENESS SCORE BETWEEN CCL AND NCCL GROUPS

Variable	CCL (N=21)			NCCL (N=15)			CCL versus NCCL
	Pre M (SD)	Post M (SD)	Pre–Post t (p)	Pre M (SD)	Post M (SD)	Pre–Post t (p)	Post t (p)
AI awareness	32.43 (2.135)	34.33 (2.652)	−4.316 (0.000)	30.20 (1.859)	32.47 (1.685)	−4.141 (0.001)	7.491 (0.000)
Perceived comfort	14.43 (1.287)	18.00 (1.265)	−11.699 (0.000)	14.73 (1.163)	15.47 (0.915)	−2.955 (0.010)	6.607 (.000)
Attitude	15.76 (1.670)	18.57 (1.660)	−6.388 (0.000)	16.00 (1.195)	16.47 (.915)	−1.388 (0.187)	4.865 (.000)

TABLE IV
COMPARISON OF CRITICAL THINKING SCORE BETWEEN CCL AND NCCL GROUPS

Variable	CCL (N=21)			NCCL (N=15)			CCL versus NCCL
	Pre M (SD)	Post M (SD)	Pre–Post t (p)	Pre M (SD)	Post M (SD)	Pre–Post t (p)	Post t (p)
Critical thinking	107.95 (11.280)	113.71 (7.869)	−3.392 (0.003)	110.07 (12.068)	123.73 (9.881)	−5.209 (0.000)	−3.386 (0.002)
Truth seeking	11.57 (0.978)	12.29 (0.902)	−4.176 (0.000)	12.00 (1.414)	14.00 (1.254)	−5.292 (0.000)	−4.779 (0.000)
Analyticity	11.71 (0.784)	12.52 (0.981)	−3.996 (0.001)	12.67 (1.799)	14.00 (1.134)	−3.251 (0.006)	−4.172 (0.000)
Systematicity	29.81 (3.544)	30.81 (2.822)	−1.296 (0.210)	30.00 (3.946)	32.53 (4.704)	−2.454 (0.028)	−1.266 (0.219)
Inquisitiveness	25.71 (4.776)	27.90 (3.097)	−2.878 (0.009)	26.80 (2.908)	30.87 (2.326)	−5.646 (0.000)	−3.123 (0.004)
Open-mindedness	29.14 (3.902)	30.19 (2.786)	−2.158 (0.043)	28.60 (4.339)	32.33 (2.257)	−3.957 (0.001)	−2.455 (0.019)

(t = –1.296, p = 0.210). In contrast, the NCCL group showed significant differences in both overall critical thinking scores and all subdimensions ($p < 0.05$), indicating a positive development in students' critical thinking during the course learning and cultural product creation process.

Independent samples t-test results revealed that the NCCL group had a significantly higher average overall critical thinking score (M = 123.73, SD = 9.881) compared to the CCL group (M = 113.71, SD = 7.869), t = –3.386, p = 0.002. These results indicated a significant difference in critical thinking outcomes between the CCL group and the NCCL group. Regarding subdimensions, truth seeking (t = –4.779, p = 0.000), analyticity (t

TABLE V
COMPARISON OF COGNITIVE LOAD SCORE BETWEEN CCL AND NCCL GROUPS

Variable	Group	Pre–Post		Post–Post			
		Z	p	Mean Rank	Ranks sum	Mann–Whitney U	p
Cognitive load	CCL (N=21)	−0.625	0.532	13.17	276.50	45.500	0.000
	NCCL (N=15)	−2.946	0.003	25.97	389.50		

$= -4.172$, $p = 0.000$), inquisitiveness ($t = -3.123$, $p = 0.004$), and open-mindedness ($t = -2.455$, $p = 0.019$) demonstrated statistically significant differences between the groups, but systematicity was not significant ($t = -1.266$, $p = 0.219$).

In addition, an analysis of chat transcripts revealed that students in the CCL group tended to overrely on ChatGPT's responses. For instance, some students identified inaccuracies in ChatGPT's responses but limited their engagement to pointing out issues and requesting regenerated information, with minimal discussion or critical evaluation of the content. Conversely, students in the NCCL group emphasized the importance of "repeated reasoning, discussion, and critical evaluation of the product's function and market value" during collaborative tasks.

D. Cognitive Load

The results of the Mann–Whitney U test revealed that cognitive load was significantly lower in the CCL group compared to the NCCL group ($U = 45.500$, $p < 0.01$; see Table V).

Analysis of chat transcripts highlights students in the CCL group actively engaged in dialogues with ChatGPT, posing questions to acquire relevant information. Common topics included cultural themes (e.g., dragon boat and theatrical culture), entrepreneurship concepts (e.g., market analysis and value promotion), and cultural product design and creation (e.g., laser cutting and programming knowledge). The assistance provided by ChatGPT's expert knowledge alleviated students' need for extensive information searches, thereby reducing cognitive effort.

In contrast, students in the NCCL group had to rely on Internet searches, literature review, and expert consultations to gather information, resulting in greater cognitive effort.

V. DISCUSSION

In the age of GAI, the role of technology inevitably changes how learners collaborate and solve problems. This study, conducted through a quasi-experimental design, explores the effects of incorporating ChatGPT into the STEM course toward entrepreneurship on students' learning performance, critical thinking, AI awareness, and cognitive load. These results highlight the complex role of AI tools in supporting entrepreneurship education, emphasizing that while they offer advantages, their potential negative impacts should also be considered.

A. H1: Introduction of ChatGPT in STEM Courses Toward Entrepreneurship Is Expected to Positively Impact on Students' Learning Performance

Regarding H1, the results of our research indicate that GAI tools, such as ChatGPT, enhance students' learning performance. These findings are consistent with the results reported in [50]. Students' tasks involve not only applying entrepreneurial knowledge but also designing and producing cultural products. Regarding novelty and fluency, multiround dialogues between group members and ChatGPT facilitated the divergence and convergence of individual thinking. As a result, students generated more innovative ideas and perspectives, which assisted them in completing cultural products. Given that ChatGPT is a large language model that supports an extensive corpus, it can simulate conversations with students, offering real-time on-demand assistance for those seeking to resolve specific concepts or problems [51]. In entrepreneurial practice, the primary selling point of cultural product design lies in its creativity, which can attract target audiences and hold commercial value. Students must apply previously acquired theoretical knowledge to develop business plans. In this process, ChatGPT serves as an "expert" to help students understand new concepts, entrepreneurial design paradigms, and interdisciplinary knowledge encountered during cultural product design and creation. For tasks involving high cognitive demand, inquiry-based learning through group collaboration allows for the sharing of relevant information and knowledge [52]. Therefore, integrating ChatGPT's API into social media platforms fosters effective interaction and collaboration between team members and intelligent machines, leading to the development of culturally innovative products with novelty and fluency.

Furthermore, the use of ChatGPT in the cultural product creation process did not significantly affect originality, suggesting that originality is an intrinsic quality that necessitates continuous knowledge accumulation and active participation [53]. As students engage in the process of identifying, analyzing, and solving problems, they refine their original skills. However, the level of originality exhibited by these tools may not fully reflect the development of an individual's originality. On the one hand, students primarily rely on STEM knowledge and technological tools to design and produce cultural products. These tasks, which involve high cognitive demands, are inherently challenging and may hinder students from generating new concepts or solving unique problems in the short term, leading to lower originality in the cultural products. On the other hand, in the early weeks of the study, students' use of ChatGPT was relatively superficial, as they primarily treated it as a basic search engine. As a result, students did not fully leverage ChatGPT's advanced questioning capabilities to explore, analyze, and solve problems, which, in turn, limited the creativity of their cultural products. Students F and Z remarked, "Although we were introduced to the use of ChatGPT at the beginning of the course, we still misused it during the process of creating cultural products, thinking it was merely an ordinary search engine." In fact, the convenience of ChatGPT in processing text-based input diminishes the originality of the work, as outcomes from complex activities involving multiple cognitive domains often lack creativity [54]. Learning demands creativity to generate novel ideas and innovations, which, in turn, provide students with actionable personalized feedback.

B. H2: Usage of ChatGPT Will Enhance AI Awareness Among Undergraduate Students

H2 indicates a significant difference in the posttest scores of AI awareness between the CCL group and the NCCL group ($p = 0.000$). This finding indicates that the advantages of GAI in convenience, timeliness, and creativity significantly facilitated the development of students' AI awareness, aligning with the findings of [55] and [56]. Students exhibit increased perceived comfort and a more positive attitude toward GAI use, believing that it can effectively improve productivity and learning outcomes. This enhancement in comfort and positive attitude allows GAI to significantly boost students' AI awareness in a short period of time. Unlike traditional AI tools, GAI enhances students' comprehension and awareness of AI technology through its real-time conversational features. For instance, in group projects, when students plan to develop electronic shadow puppetry commercial products, they can use ChatGPT to compare their product with existing market products and receive valuable suggestions. In addition, ChatGPT simulates Socratic-style reasoning from a first-principles perspective, helping students distill key insights from complex business information [57]. These interactions not only enhance students' instrumental understanding of AI but also help them progressively realize the potential and limitations of AI technology, especially in its application to information processing and problem solving. However, analysis of chat records and learning reflections reveals that the enhancement of students' AI awareness when using GAI is primarily focused on their instrumental understanding. Specifically, students view GAI as a search engine for retrieving information, rather than as a partner, suggesting that they have not yet made the transition from instrumental to relational awareness. This phenomenon is consistent with the technology acceptance model, indicating that students' acceptance of new technologies necessitates time and repeated usage experiences. As GAI continues to be used, students' AI awareness progressively shifts from understanding the tool itself to considering AI ethics and potential application risks. Research suggests that the more mature and stable awareness becomes, the more it facilitates and guides rational behavior [58]. In this study, students identified issues, such as privacy breaches, data security, and algorithmic bias, during repeated Q&A interactions, which prompted them to avoid or minimize these negative effects in human–machine interactions.

The development of AI awareness is a gradual process, particularly manifested in the subdimensions of perceived comfort and attitude. Studies have shown that students' perceived comfort with AI tools correlates positively with their confidence and willingness to use ChatGPT in learning [59]. In the early stages, students tend to overlook the limitations of AI and rely primarily on its speed to complete tasks. As students gain proficiency in using ChatGPT, its reliability and consistency improve through repeated interactions, which, in turn, further enhances their perceived comfort. Moreover, the effectiveness of ChatGPT in learning is progressively validated, and students' perceived comfort and intent to use it also significantly increase. For instance, as a "thought accelerator," ChatGPT not only provides solutions but also fosters creativity, helping students generate a variety of responses, which significantly enhances their intention to use the tool. Second, attitude refers to the level of interest and positive evaluation individuals have toward the use of a specific technology [60]. The ease of use and usefulness of GAI are critical in shaping students' attitudes toward its use. Through interactions with ChatGPT, students come to recognize the tool's efficiency, particularly when faced with complex learning materials, as ChatGPT helps them rapidly extract and summarize relevant information. As a result, students' expectations and intentions to use ChatGPT increase with its growing ease of use and usefulness, leading to higher learning motivation and engagement. Interestingly, in contrast to the CCL group, the increase in AI awareness for the NCCL group was relatively modest. Some students, having heard about ChatGPT's "magical powers" from their peers, engaged with the tool out of curiosity but did not deeply utilize it to design and create cultural products. Consequently, their perceived comfort and attitude improved to some extent, but the increase was minimal. This phenomenon suggests that mere curiosity or interest is not enough to significantly boost students' AI awareness. Only through repeated multiturn interactions with ChatGPT, coupled with the gradual refinement of usage strategies (such as adjusting prompts and employing BERT instructions), can students truly grasp the potential of AI tools.

C. H3: Applying ChatGPT in Collaborative Learning Activities Does Not Effectively Cultivate Students' Critical Thinking Skills

Interestingly, the results of the posttest independent samples t-test revealed that the critical thinking scores of the CCL group were significantly lower than those of the NCCL group ($p = 0.000$), and H3 was not confirmed. This finding is consistent with the research of Fuchs [40], which suggests that overreliance on large language models like ChatGPT can hinder the development of critical thinking skills. In the STEM course toward entrepreneurship, students tend to become passive learners, continuously absorbing information from these large language models without critically assessing the accuracy and relevance of the provided information. Specifically, as ChatGPT improves, the information, answers, or strategies it provides increasingly align with standard solutions [61], which may lead students to forgo independent thinking. This stands in sparking contrast to the findings of Emmert-Streib [62] and Cooper [63], who argue that the use of ChatGPT can enhance students' critical thinking. Despite the fact that teaching assistants had provided students with guidance on how to use ChatGPT before the experiment, they still viewed it as a simple search engine for retrieving and gathering relevant information, with minimal use of prompts to formulate thoughtful questions.

Due to the inherent limitations of chatbots' language models, ChatGPT often generates misleading and confusing information. For instance, when dealing with content related to local Chinese culture, ChatGPT fails to provide innovative insights, with much of the content appearing fabricated, which diminishes students' willingness to engage in critical reflection. This issue was identified in week 12, at which point the teacher highlighted the deficiencies of ChatGPT-generated answers using several practical examples and encouraged students to engage in critical reflection on the provided answers. In the subdimensions of critical thinking, the posttest independent samples t-test did not show significant results for systematicity. The development of systematic thinking requires prolonged training and multiple practices. An eight-week experimental period might be insufficient to fully leverage the potential of ChatGPT. Moreover, both the CCL and NCCL groups needed to consider the materials, structure, and functionality of the cultural products during the creation process, which might not have been significantly influenced by ChatGPT.

In addition, the results of the paired samples t-test for the CCL and NCCL groups revealed significant differences in critical thinking scores between the pretest and posttest ($p < 0.05$). This contradicts existing research conclusions, which argue that critical thinking cannot be significantly improved in a short period [64]. In this study, the course content includes various entrepreneurial concepts (such as opportunity recognition and minimum viable products) and technical knowledge (such as programming and 3-D modeling), which require students to repeatedly hypothesize, test, and refine solutions during both knowledge learning and product development, in order to evaluate their feasibility. This indicates that the abstract nature and difficulty of the course content play a role in stimulating students' critical thinking [65], which could be a key factor influencing the experimental results. Moreover, the STEM entrepreneurship course incorporates a variety of problem-solving activities, including reasoning, evidence evaluation, and logical analysis, all of which can significantly strengthen students' critical thinking in the short term [66]. Furthermore, as the NCCL group did not have access to ChatGPT, they were more inclined to use high-level journal articles, Baidu, and Viki tools to cross-check the validity of their ideas and seek optimal solutions, which could account for their relatively higher critical thinking abilities.

Conversely, the CCL group used ChatGPT to offload cognitive resources in order to analyze and solve complex problems, potentially giving students more time and energy to engage in critical thinking. Specifically, ChatGPT provides multiple perspectives on the same topic, encouraging students to challenge their current viewpoints and continuously reflect on their decisions and assumptions, which could enhance students' critical thinking. In addition, reflection reports revealed that some students recognized the inaccuracies or errors in the information provided by ChatGPT, prompting them to critically review the information and seek validation from other sources, which could foster their critical thinking. However, the passive use of ChatGPT may foster cognitive dependence, potentially replacing the thinking process, which could explain why the CCL group had significantly lower posttest scores than the NCCL group. It is important to emphasize that in this study, the course content and the design of learning activities may have a more substantial impact on critical thinking than the influence of ChatGPT.

D. H4: The Integration of ChatGPT Will Optimize Cognitive Load, Further Enhancing the Learning Efficiency of Students in Entrepreneurship Practice Activities

The posttest cognitive load scores of the CCL group were lower than their pretest scores, whereas the cognitive load scores of the NCCL group significantly increased. In addition, the posttest independent samples t-test revealed a significant difference between the two groups. This suggests that ChatGPT's assistance in completing tasks and solving ill-structured problems can effectively reduce cognitive load, aligning with existing research findings [67]. Given the limited capacity of human working memory to process complex problems, which can constrain the exploration of interdisciplinary issues [49], the application of ChatGPT—capable of perception, generation, and creativity—to interpret students' linguistic variations in a human-like manner and respond appropriately, can substantially reduce the mental effort students invest in creating cultural products and comprehending STEM and entrepreneurship concepts. Although ChatGPT does not always provide entirely accurate answers, its rapid responses and timely feedback can facilitate the divergence and convergence of students' thinking, aiding them in understanding STEM, entrepreneurship, and cultural knowledge, reducing cognitive load, and enhancing problem-solving efficiency.

Furthermore, while ChatGPT demonstrates proficiency in handling common-sense questions, its corpus lacks training in native Chinese culture (e.g., drama, folk music, geography, festivals, etc.) [68], resulting in less diverse or lower quality training data and occasionally inaccurate responses, positioning it as an inexperienced problem solver. As a result, students often need to allocate additional time and effort to literature reviews and expert consultations, which increases their cognitive load. Moreover, the primary objective of this course is to enable students to apply STEM and entrepreneurship knowledge in entrepreneurial practice activities, facilitating learning and inquiry among group members through complex and authentic entrepreneurial problems or concepts. Since complex and abstract entrepreneurial problems or STEM concepts may demand additional cognitive resources [69], students with limited prior knowledge of STEM and entrepreneurship are more likely to encounter challenges in problem solving and solution design, leading to high levels of cognitive load for both groups. Consequently, maintaining an appropriate level of cognitive load in entrepreneurship practice activities is essential to prevent students from becoming overly reliant on ChatGPT as a tool, ensuring that they use intelligent machines effectively to solve problems and improve learning outcomes.

VI. Conclusion and Implication

Technologies inherently present dual possibilities, offering both opportunities and challenges. While transformative technologies create opportunities for STEM and entrepreneurship education, they also pose potential risks. The study's findings of this advance the theoretical frameworks of synergetic theory and distributed cognition theory. The advent of ChatGPT has transformed traditional proxy technological tools into intelligent human-like machines. Integrating ChatGPT into multiuser multimachine collaborative learning environments deepens our understanding of its applications and inherent risks in complex interactive contexts. Furthermore, this research contributes to the practical applications and measurable impacts of ChatGPT in STEM entrepreneurship education. The results indicate that real-time interaction and innovative feedback substantially improve various aspects of student learning, including learning performance, AI awareness, and cognitive load. However, it is crucial to acknowledge the inherent limitations of the technology, especially its tendency to inhibit critical thinking skills. Moreover, tools like GAI hold promise for educational innovation and curriculum development, with this study offering actionable insights for educator in designing and facilitating learning experiences.

GAI, exemplified by ChatGPT, functions as an "external brain," seamlessly integrating into cognitive process and extending the capabilities of human cognition. ChatGPT enhances the completeness and creativity of student learning outcomes by automating routine tasks, stimulating critical thinking, and offering tailored guidance. The findings of this study indicate that ChatGPT's user-friendliness, high interactivity, and extensive database of ChatGPT, demonstrating its effectiveness in aiding students' acquisition of complex STEM and entrepreneurship knowledge, improving AI awareness, boosting creativity, and alleviating cognitive load. In addition, ChatGPT excels in addressing theoretical inquiries and expediting both divergent and convergent thinking processes during problem-solving activities. However, it is essential to recognize that entrepreneurship requires the ability to identify business opportunities and make informed decisions grounded in current international political and economic contexts. However, ChatGPT's database may lack the most current entrepreneurial information, potentially hindering students' ability to make informed decisions. Integrating ChatGPT's API with social media platforms facilitates iterative dialogues and collaboration between diverse learners and ChatGPT, leveraging collective intelligence to support the application of interdisciplinary knowledge in addressing complex problems and creating cultural products.

Furthermore, it is essential that technology acts as a key driver in shaping individuals' technological behaviors. The rationale for integrating technology into education lies in aligning student needs with the capabilities of the tools used. Thus, when integrating ChatGPT into STEM course toward entrepreneurship, it is crucial to consider the psychological adjustment and significant shifts in students' demands. Moreover, efforts should focus on optimizing ChatGPT to serve as a more effective learning tool. First, guide students in mastering various information retrieval channels and effective search methods, enabling them to obtain relevant information that reduces cognitive load. This, in turn, allows students to focus on higher-level creative tasks, ultimately generating more valuable products. Second, use appropriate instructional scaffolding to encourage students to compare information from different sources, critically evaluate its logic and authenticity, and foster the development of their critical thinking skills. As ChatGPT lacks a nuanced understanding of domain-specific knowledge (particularly cultural knowledge), it may generate feedback that lacks depth and insight. Third, ChatGPT's instrumental rationality can diminish individuals' sense of agency and reflective thinking. Caution should be taken when using informational tools in situations of cognitive conflict (where ChatGPT's information conflicts with students' preexisting knowledge, causing cognitive confusion), and individuals should be guided to use technology in a way that preserves their autonomy. Fourth, the database underlying ChatGPT presents challenges in preventing the misuse or abuse of educational data during processes such as collection, storage, analysis, and sharing. Therefore, both educators and students must actively improve their AI literacy and ensure that ChatGPT is used ethically within educational contexts.

Finally, due to the teaching team's lack of prior experience with ChatGPT, students were not initially made aware of the associated risks and ethical concerns, and the absence of timely follow-up and supervision lead to misuse of ChatGPT, potentially affecting the experimental results. Analysis of student WeChat group chat records revealed that some business plans and cultural products were heavily influenced by ChatGPT, lacking original thinking and insights. This phenomenon raises concerns regarding academic integrity, which requires close attention from educators. In addition, it is crucial to prevent students from using the tool for entertainment purposes beyond its educational role. Some students initially treated ChatGPT as an entertainment tool but, after their initial interest waned, began to use it merely as a search engine. As the enhancement of AI awareness is a gradual process, it is essential to foster the transition from instrumental to relational cognition through repeated interactions and practical applications, thereby unlocking the multidimensional potential of GAI. Moreover, the results of our study may be influenced by the limited sample size, which consisted only of undergraduate students in educational technology, and the relatively short duration of CCL. Future research should incorporate longer experimental periods, more adaptive learning scaffolds, and strategies to foster collaborative learning, ultimately advancing the vision of human–machine cocreation and blended learning.

References

[1] S. Kaya-Capocci, "The trends in entrepreneurship and STEM education studies: A bibliometric study," *İnönü Üniversitesi Eğitim Fakültesi Dergisi*, vol. 24, no. 2, pp. 869–892, Sep. 2023, doi: 10.17679/inuefd.1320031.

[2] X. Weng, T. K. F. Chiu, and C. C. Tsang, "Promoting student creativity and entrepreneurship through real-world problem-based maker education," *Thinking Skills Creativity*, vol. 45, Sep. 2022, Art. no. 101046, doi: 10.1016/j.tsc.2022.101046.

[3] M. G. Colombo and E. Piva, "Start-ups launched by recent STEM university graduates: The impact of university education on entrepreneurial entry," *Res. Policy*, vol. 49, no. 6, Jul. 2020, Art. no. 103993, doi: 10.1016/j.respol.2020.103993.

[4] M. Eltanahy, S. Forawi, and N. Mansour, "Incorporating entrepreneurial practices into STEM education: Development of interdisciplinary E-STEM model in high school in the United Arab Emirates," *Thinking Skills Creativity*, vol. 37, Sep. 2020, Art. no. 100697, doi: 10.1016/j.tsc.2020.100697.

[5] S. Primario, P. Rippa, and G. Secundo, "Rethinking entrepreneurial education: The role of digital technologies to assess entrepreneurial self-efficacy and intention of STEM students," *IEEE Trans. Eng. Manage.*, vol. 71, pp. 2829–2842, 2024, doi: 10.1109/TEM.2022.3199709.

[6] U. Sari, H. Çelik, H. M. Pektaş, and S. Yalçın, "Effects of STEM-focused Arduino practical activities on problem-solving and entrepreneurship skills," *Australas. J. Educ. Technol.*, vol. 38, no. 3, pp. 135–149, May 2022, doi: 10.14742/ajet.7293.

[7] K. I. Roumeliotis and N. D. Tselikas, "ChatGPT and open-AI models: A preliminary review," *Future Internet*, vol. 15, no. 6, May 2023, Art. no. 192, doi: 10.3390/fi15060192.

[8] I. Cribben and Y. Zeinali, "The benefits and limitations of ChatGPT in business education and research: A focus on management science, operations management and data analytics," *SSRN Electron. J.*, 2023, doi: 10.2139/ssrn.4404276.

[9] M. Vecchiarini and T. Somià, "Redefining entrepreneurship education in the age of artificial intelligence: An explorative analysis," *Int. J. Manage. Educ.*, vol. 21, no. 3, Nov. 2023, Art. no. 100879, doi: 10.1016/j.ijme.2023.100879.

[10] Y. Ji, X. Zou, T. Li, and Z. Zhan, "The effectiveness of ChatGPT on pre-service teachers' STEM teaching literacy, learning performance, and cognitive load in a teacher training course," in *Proc. 6th Int. Conf. Educ. Technol. Manage.*, Guangzhou, China, 2023, pp. 16–22, doi: 10.1145/3637907.3637948.

[11] T. Li, Y. Ji, and Z. Zhan, "Expert or machine? Comparing the effect of pairing student teacher with in-service teacher and ChatGPT on their critical thinking, learning performance, and cognitive load in an integrated-STEM course," *Asia Pacific J. Educ.*, vol. 44, no. 1, pp. 45–60, Jan. 2024, doi: 10.1080/02188791.2024.2305163.

[12] H. Miço and J. Cungu, "Entrepreneurship education, a challenging learning process towards entrepreneurial competence in education," *Administ. Sci.*, vol. 13, no. 1, Jan. 2023, Art. no. 22, doi: 10.3390/admsci13010022.

[13] D. BaiDoo-Anu and L. O. Ansah, "Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning," *J. AI*, vol. 7, no. 1, pp. 52–62, Dec. 2023, doi: 10.61969/jai.1337500.

[14] A. Kelly, M. Sullivan, and K. Strampel, "Generative artificial intelligence: University student awareness, experience, and confidence in use across disciplines," *J. Univ. Teach. Learn. Pract.*, vol. 20, no. 6, Aug. 2023, doi: 10.53761/1.20.6.12.

[15] B. M. Lake, T. D. Ullman, J. B. Tenenbaum, and S. J. Gershman, "Building machines that learn and think like people," *Behav. Brain Sci.*, vol. 40, 2017, Art. no. e253, doi: 10.1017/S0140525X16001837.

[16] A. Brailas, "Postdigital duoethnography: An inquiry into human-artificial intelligence synergies," *Postdigital Sci. Educ.*, vol. 6, no. 2, pp. 486–515, Jun. 2024, doi: 10.1007/s42438-024-00455-7.

[17] J. Hollan, E. Hutchins, and D. Kirsh, "Distributed cognition: Toward a new foundation for human-computer interaction research," *ACM Trans. Comput.-Hum. Interact.*, vol. 7, no. 2, pp. 174–196, Jun. 2000, doi: 10.1145/353485.353487.

[18] A. J. C. Sharkey, "Should we welcome robot teachers?," *Ethics Inf. Technol.*, vol. 18, no. 4, pp. 283–297, Dec. 2016, doi: 10.1007/s10676-016-9387-z.

[19] L. Chen, P. Chen, and Z. Lin, "Artificial intelligence in education: A review," *IEEE Access*, vol. 8, pp. 75264–75278, 2020, doi: 10.1109/ACCESS.2020.2988510.

[20] X. Deng and Z. Yu, "A meta-analysis and systematic review of the effect of chatbot technology use in sustainable education," *Sustainability*, vol. 15, no. 4, Feb. 2023, Art. no. 2940, doi: 10.3390/su15042940.

[21] T. Susi and T. Ziemke, "Social cognition, artefacts, and stigmergy: A comparative analysis of theoretical frameworks for the understanding of artefact-mediated collaborative activity," *Cogn. Syst. Res.*, vol. 2, no. 4, pp. 273–290, Dec. 2001, doi: 10.1016/S1389-0417(01)00053-5.

[22] A. C. Graesser, S. M. Fiore, S. Greiff, J. Andrews-Todd, P. W. Foltz, and F. W. Hesse, "Advancing the science of collaborative problem solving," *Psychol. Sci. Public Int.*, vol. 19, no. 2, pp. 59–92, Nov. 2018, doi: 10.1177/1529100618808244.

[23] N. J. Kim, B. R. Belland, and A. E. Walker, "Effectiveness of computer-based scaffolding in the context of problem-based learning for stem education: Bayesian meta-analysis," *Educ. Psychol. Rev.*, vol. 30, no. 2, pp. 397–429, Jun. 2018, doi: 10.1007/s10648-017-9419-1.

[24] K. Guo and D. Wang, "To resist it or to embrace it? Examining ChatGPT's potential to support teacher feedback in EFL writing," *Educ. Inf. Technol.*, vol. 29, no. 7, pp. 8435–8463, May 2024, doi: 10.1007/s10639-023-12146-0.

[25] Q. Xia, T. K. F. Chiu, M. Lee, I. T. Sanusi, Y. Dai, and C. S. Chai, "A self-determination theory (SDT) design approach for inclusive and diverse artificial intelligence (AI) education," *Comput. Educ.*, vol. 189, Nov. 2022, Art. no. 104582, doi: 10.1016/j.compedu.2022.104582.

[26] T. Shaik et al., "A review of the trends and challenges in adopting natural language processing methods for education feedback analysis," *IEEE Access*, vol. 10, pp. 56720–56739, 2022, doi: 10.1109/ACCESS.2022.3177752.

[27] L. Luan, X. Lin, Y. Dai, S. Hu, and Q. Sun, "Exploring the cognitive dynamics of artificial intelligence in the post-COVID-19 and Learning 3.0 era: A case study of ChatGPT," *AsianJDE*, vol. 19, no. 2, pp. 1–10, Apr. 2024, doi: 10.5281/zenodo.10990472.

[28] Y. Su and Z. Liu, "A study of ChatGPT empowering college students' innovation and entrepreneurship education in the context of China," *Int. J. New Develop. Educ.*, vol. 5, no. 13, pp. 1–7, 2023, doi: 10.25236/IJNDE.2023.051301.

[29] T. McCaffrey and L. Spector, "An approach to human–machine collaboration in innovation," *Artif. Intell. Eng. Des. Anal. Manuf.*, vol. 32, no. 1, pp. 1–15, Feb. 2018, doi: 10.1017/S0890060416000524.

[30] A. O. Ajlouni, F. A.-A. Wahba, and A. Salem Almahaireh, "Students' attitudes towards using ChatGPT as a learning tool: The case of the University of Jordan," *Int. J. Interact. Mobile Technol.*, vol. 17, no. 18, pp. 99–117, Sep. 2023, doi: 10.3991/ijim.v17i18.41753.

[31] P. P. Ray, "ChatGPT: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope," *Internet Things Cyber-Phys. Syst.*, vol. 3, pp. 121–154, 2023, doi: 10.1016/j.iotcps.2023.04.003.

[32] T. Hirosawa, Y. Harada, M. Yokose, T. Sakamoto, R. Kawamura, and T. Shimizu, "Diagnostic accuracy of differential-diagnosis lists generated by generative pretrained transformer 3 chatbot for clinical vignettes with common chief complaints: A pilot study," *Int. J. Environ. Res. Public Health*, vol. 20, no. 4, Feb. 2023, Art. no. 3378, doi: 10.3390/ijerph20043378.

[33] J. Rudolph, S. Tan, and S. Tan, "ChatGPT: Bullshit spewer or the end of traditional assessments in higher education?," *J. Appl. Learn. Teach.*, vol. 6, no. 1, pp. 342–363, Jan. 2023, doi: 10.37074/jalt.2023.6.1.9.

[34] M. Farrokhnia, S. K. Banihashem, O. Noroozi, and A. Wals, "A SWOT analysis of ChatGPT: Implications for educational practice and research," *Innov. Educ. Teach. Int.*, vol. 61, no. 3, pp. 460–474, May 2024, doi: 10.1080/14703297.2023.2195846.

[35] C. Winkler, B. Hammoda, E. Noyes, and M. Van Gelderen, "Entrepreneurship education at the dawn of generative artificial intelligence," *Entrepreneurship Educ. Pedagogy*, vol. 6, no. 4, pp. 579–589, Oct. 2023, doi: 10.1177/25151274231198799.

[36] B. Hammoda, "ChatGPT for founding teams: An entrepreneurial pedagogical innovation," *Int. J. Technol. Educ.*, vol. 7, no. 1, pp. 154–173, Feb. 2024, doi: 10.46328/ijte.530.

[37] R. K. Goel and M. A. Nelson, "Awareness of artificial intelligence: Diffusion of AI versus ChatGPT information with implications for entrepreneurship," *J. Technol. Transfer*, vol. 50, no. 1, pp. 96–113, Feb. 2025, doi: 10.1007/s10961-024-10089-3.

[38] C. Prentice, S. Weaven, and I. A. Wong, "Linking AI quality performance and customer engagement: The moderating effect of AI preference," *Int. J. Hospitality Manage.*, vol. 90, Sep. 2020, Art. no. 102629, doi: 10.1016/j.ijhm.2020.102629.

[39] D. R. E. Cotton, P. A. Cotton, and J. R. Shipway, "Chatting and cheating: Ensuring academic integrity in the era of ChatGPT," *Innov. Educ. Teach. Int.*, vol. 61, no. 2, pp. 228–239, Mar. 2023, doi: 10.1080/14703297.2023.2190148.

[40] K. Fuchs, "Exploring the opportunities and challenges of NLP models in higher education: Is ChatGPT a blessing or a curse?," *Front. Educ.*, vol. 8, May 2023, Art. no. 1166682, doi: 10.3389/feduc.2023.1166682.

[41] L. Skavronskaya, A. (Hana) Hadinejad, and D. Cotterell, "Reversing the threat of artificial intelligence to opportunity: A discussion of ChatGPT in tourism education," *J. Teach. Travel Tourism*, vol. 23, no. 2, pp. 253–258, Apr. 2023, doi: 10.1080/15313220.2023.2196658.

[42] T.-T. Wu, H.-Y. Lee, P.-H. Li, C.-N. Huang, and Y.-M. Huang, "Promoting self-regulation progress and knowledge construction in blended learning via ChatGPT-based learning aid," *J. Educ. Comput. Res.*, vol. 61, no. 8, pp. 3–31, Jan. 2024, doi: 10.1177/07356331231191125.

[43] M. Edwards-Schachter, A. García-Granero, M. Sánchez-Barrioluengo, H. Quesada-Pineda, and N. Amara, "Disentangling competences: Interrelationships on creativity, innovation and entrepreneurship," *Thinking Skills Creativity*, vol. 16, pp. 27–39, Jun. 2015, doi: 10.1016/j.tsc.2014.11.006.

[44] T. M. Amabile, "Social psychology of creativity: A consensual assessment technique," *J. Pers. Soc. Psychol.*, vol. 43, no. 5, pp. 997–1013, Nov. 1982, doi: 10.1037/0022-3514.43.5.997.

[45] T. R. Kurtzberg, "Feeling creative, being creative: An empirical study of diversity and creativity in teams," *Creativity Res. J.*, vol. 17, no. 1, pp. 51–65, Feb. 2005, doi: 10.1207/s15326934crj1701_5.

[46] C. Chen, "An experimental study about implementing the teaching strategies of critical thinking in social class of 5th grade elementary school," M.S. thesis, Dept. Educ., Nat. Chengchi Univ., Taipei, Taiwan, 2000.

[47] F. G. W. C. Paas and J. J. G. Van Merriënboer, "Instructional control of cognitive load in the training of complex cognitive tasks," *Educ. Psychol. Rev.*, vol. 6, no. 4, pp. 351–371, Dec. 1994, doi: 10.1007/BF02213420.

[48] D. Gopher and R. Braune, "On the psychophysics of workload: Why bother with subjective measures?," *Hum. Factors*, vol. 26, no. 5, pp. 519–532, Oct. 1984, doi: 10.1177/001872088402600504.

[49] F. Paas, J. E. Tuovinen, H. Tabbers, and P. W. M. Van Gerven, "Cognitive load measurement as a means to advance cognitive load theory," *Educ. Psychol.*, vol. 38, no. 1, pp. 63–71, Jan. 2003, doi: 10.1207/S15326985EP3801_8.

[50] S. Alneyadi and Y. Wardat, "ChatGPT: Revolutionizing student achievement in the electronic magnetism unit for eleventh-grade students in Emirates schools," *Contemp. Educ. Technol.*, vol. 15, no. 4, Oct. 2023, Art. no. ep448, doi: 10.30935/cedtech/13417.

[51] A. Gilson et al., "How does ChatGPT perform on the United States medical licensing examination? The implications of large language models for medical education and knowledge assessment," *JMIR Med. Educ.*, vol. 9, Feb. 2023, Art. no. e45312, doi: 10.2196/45312.

[52] B. Zhong and T. Li, "Can pair learning improve students' troubleshooting performance in robotics education?," *J. Educ. Comput. Res.*, vol. 58, no. 1, pp. 220–248, Mar. 2020, doi: 10.1177/0735633119829191.

[53] K. S. Hyatt, "Creativity through intrapersonal communication dialog," *J. Creative Behav.*, vol. 26, no. 1, pp. 65–71, Mar. 1992, doi: 10.1002/j.2162-6057.1992.tb01158.x.

[54] Y. K. Dwivedi et al., "Opinion paper: 'so what if ChatGPT wrote it?' Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy," *Int. J. Inf. Manage.*, vol. 71, Aug. 2023, Art. no. 102642, doi: 10.1016/j.ijinfomgt.2023.102642.

[55] G. Liu and C. Ma, "Measuring EFL learners' use of ChatGPT in informal digital learning of English based on the technology acceptance model," *Innov. Lang. Learn. Teach.*, vol. vol. 18, no. 2, pp. 125–138, Mar. 2024, doi: 10.1080/17501229.2023.2240316.

[56] J. Belda-Medina and J. R. Calvo-Ferrer, "Using chatbots as AI conversational partners in language learning," *Appl. Sci.*, vol. 12, no. 17, Aug. 2022, Art. no. 8427, doi: 10.3390/app12178427.

[57] A. Balmer, "A sociological conversation with ChatGPT about AI ethics, affect and reflexivity," *Sociology*, vol. 57, no. 5, pp. 1249–1258, May 2023, doi: 10.1177/00380385231169676.

[58] H. M. N. AlKanaan, "Awareness regarding the implication of artificial intelligence in science education among pre-service science teachers," *Int. J. Instruct.*, vol. 15, no. 3, pp. 895–912, Jul. 2022, doi: 10.29333/iji.2022.15348a.

[59] S. Ebadi and A. Amini, "Examining the roles of social presence and human-likeness on Iranian EFL learners' motivation using artificial intelligence technology: A case of CSIEC chatbot," *Interact. Learn. Environ.*, vol. 32, no. 2, pp. 655–673, Feb. 2024, doi: 10.1080/10494820.2022.2096638.

[60] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology," *MIS Quart.*, vol. 13, no. 3, pp. 319–340, Sep. 1989, doi: 10.2307/249008.

[61] C. Leiter et al., "ChatGPT: A meta-analysis after 2.5 months," *Mach. Learn. Appl.*, vol. 16, Jun. 2024, Art. no. 100541, doi: 10.1016/j.mlwa.2024.100541.

[62] F. Emmert-Streib, "Importance of critical thinking to understand ChatGPT," *Eur. J. Hum. Genet.*, vol. 32, no. 4, pp. 377–378, Apr. 2024, doi: 10.1038/s41431-023-01443-4.

[63] G. Cooper, "Examining science education in ChatGPT: An exploratory study of generative artificial intelligence," *J. Sci. Educ. Technol.*, vol. 32, no. 3, pp. 444–452, Jun. 2023, doi: 10.1007/s10956-023-10039-y.

[64] Y. C. Yang and H. Chou, "Beyond critical thinking skills: Investigating the relationship between critical thinking skills and dispositions through different online instructional strategies," *Brit. J. Educ. Technol.*, vol. 39, no. 4, pp. 666–684, Jul. 2008, doi: 10.1111/j.1467-8535.2007.00767.x.

[65] O. A. Oyewo, S. Ramaila, and L. Mavuru, "Harnessing project-based learning to enhance STEM students' critical thinking skills using water treatment activity," *Educ. Sci.*, vol. 12, no. 11, Nov. 2022, Art. no. 780, doi: 10.3390/educsci12110780.

[66] R. H. Ennis, "Critical thinking and subject specificity: Clarification and needed research," *Educ. Res.*, vol. 18, no. 3, pp. 4–10, Apr. 1989, doi: 10.3102/0013189X018003004.

[67] S. Das, "Reader response: Neurology education in 2035: The neurology future forecasting series," *Neurology*, vol. 100, no. 23, pp. 1119–1120, Jun. 2023, doi: 10.1212/WNL.0000000000207441.

[68] D. H. R. Spennemann, "ChatGPT and the generation of digitally born 'knowledge': How does a generative AI language model interpret cultural heritage values?," *Knowledge*, vol. 3, no. 3, pp. 480–512, Sep. 2023, doi: 10.3390/knowledge3030032.

[69] U. Gupta and R. Z. Zheng, "Cognitive load in solving mathematics problems: Validating the role of motivation and the interaction among prior knowledge, worked examples, and task difficulty," *Eur. J. STEM Educ.*, vol. 5, no. 1, Nov. 2020, Art. no. 05, doi: 10.20897/ejsteme/9252.

Yu Ji was born in 1996. He is working toward the Ph.D. degree in educational technology with the School of Information Technology in Education, South China Normal University, Guangzhou, China.

His research interests include the application of artificial intelligence in education, STEM and entrepreneurship education, and computer-supported teaching and learning.

Zehui Zhan was born in 1983. She received the Ph.D. degree in education technology from South China Normal University, Guangzhou, China, in 2010.

She is currently the Outstanding Professor with the School of Information Technology in Education, South China Normal University, where she is also the Youth Pearl River Scholar, Hong Kong Scholar, and Principal Investigator of the Smart Educational Equipment Industry–University–Research Cooperation Base, the Associate Director of the Key Laboratory of Artificial Intelligence and Smart Education, and the Associate Director of the Guangdong Educational Evaluation Big Data Engineering Technology Research Center. She has authored or coauthored more than 120 papers in the field. Her research interests include STEM education and learning science.

Dr. Zhan is the recipient of the Annual Youth Excellent University Teacher Award from the Fok Yingdong Education Foundation and the Ministry of Education in China, the title "Best Teacher" in South China Normal University, and the first prize in the National Education Software Competition. Her doctoral dissertation was nominated as the national 100 excellent doctoral dissertations in China.

Tingting Li was born in 1992. She is working toward the Ph.D. degree in educational technology with the School of Information Technology in Education, South China Normal University, Guangzhou, China.

Her research interests include STEM education and teacher education.

Xuanxuan Zou was born in 2000. She is working toward the master's degree in modern educational technology with the School of Information Technology in Education, South China Normal University, Guangzhou, China.

She has authored or coauthored several papers in core journals. Her research interests include computational thinking and interdisciplinary teaching.

Ms. Zou received the second prize at the provincial level in the Science and Technology Labor Innovation Competition.

Siyuan Lyu was born in 2000. She is working toward the Ph.D. degree in educational technology with the School of Information Technology in Education, South China Normal University, Guangzhou, China.

Her research interests include STEM education, teacher education, and artificial intelligence education.

Ms. Lyu received the Young Scholar Award at the Asian Student Roundtable Seminar in 2022 and the third prize for excellent conference paper at the Global Chinese Conference on Inquiry Learning (GCCIL) 2021.